MIL-C-60641(MU)
15 September 1969
SUPERSEDING
FA-PD-MI-2559, Rev 0
25 October 1962

MILITARY SPECIFICATION
CARTRIDGE, DELAY, M174 ASSEMBLY

1. SCOPE

1.1 This specification covers Cartridge, Delay, M174 Assembly, for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-60539	- Coating, Anodic Conventional for Aluminum and Aluminum Alloys
MS28741	- Hose Assembly, Detachable End Fitting, Medium Pressure

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O-rings

DRAWINGS

Munitions Command

D10522656	- Cartridge, Delay, M174 Assembly

FSC 1377

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

DRAWINGS (Cont'd)

C8864168	- Carton, Inner, Packing, Ammunition for Cartridge, Delay, M174
C9214156	- Box, Packing, Ammunition for Cartridge Delay, M174
IEL10522656	- Inspection Equipment List for Cartridge, Delay, M174 Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y. 10016.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Protective finish.- Protective finish shall be in accordance with MIL-C-60539.

3.3 Vibration.- The assembled cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the delay charge, primer, igniter or propellant or derangement of components.

3.4 Primer sensitivity.- The primer shall fire when drop tested at an energy level of 26 inch-ounces in a sectioned cartridge, void of delay, igniter and propellant charges.

3.5 Functional

3.5.1 The cartridge shall not fail to fire.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

3.5.2 Ballistics.- The cartridge when tested in an M88 Initiator, shall comply with the requirements specified in Table I.

Table I

Temperature	Pressure (psi) 1/ Min.	Max.	Delay Seconds
-65° to -70°F	500	1500	2.45 max.
65° to 75°F	575	1500	2.0 ± .2
160° to 165°F	625	1500	1.6 min.

1/ Pressure values shall be produced in a 1 cubic inch pressure station at the end of 15 feet of aircraft hose, medium pressure, size no. 4.

3.6 Workmanship.- Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects.- The metal part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.6.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 23 units for protective finish tests, 50 units for hardness test, 50 for primer sensitivity, 12 for vibration (to be selected from ballistic sample) 5 for sealant tests and 30 for ballistic tests and tested in accordance with Table II and 10 sets of unassembled components (including explosive components) for dimensional and visual examination.

Table II

Paragraph Requirement	Paragraph Test	Test	Sample Size	Type of Test Destr.	Type of Test Non-Destr.
3.2	4.4.1	Protective finish	23	X	-
Dwg.	4.4.2	Hardness	50	-	X
Dwg.	4.4.3	Airtightness	each	-	X
3.3	4.4.4	Vibration	12(fr ballistic sample)	-	X
3.4	4.4.5	Primer sensitivity	50	X	-
Dwg.	4.4.6	Sealant App.	5	X	-
3.5.2	4.4.7	Ballistics	30	X	-

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and tests.- After examination and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.-The sample for ballistic tests shall be temperature conditioned and conform to requirements as indicated in Table I and III.

Table III

Temperature Degrees(°) Fahrenheit(°)	Vibrated	Not Vibrated
-65 to -70	4	6
65 to 75	4	6
160 to 165	4	6

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.2.1.3 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.- An ammunition lot shall consist of an identifiable group of metal parts assemblies consisting of:

a. Components produced by a homogeneous process.
b. Propellant from not more than one lot.
c. Primers from not more than one lot.
d. Igniters from not more than one lot.
e. Delay elements from not more than one lot.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables IV and V. For an individual defect (single characteristic) the criteria in Table IV shall be used. For total defects within a major or minor class the criteria in Table V shall be used. All non-conforming material shall be rejected. The roughness comparison specimens prescribed in ASA B46.1 shall be used as a basis of comparison for surface roughness determination. Inspection for rubber o-rings shall be in accordance with MIL-STD-413. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

Table IV

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

Table V

Sample Size for Major Defects Number of defects in class	Total allowable defects per sample size 110	150	225	300
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over 43	6	8	11	14

Sample Size for Major Defects Number of defects in class	110	150	225	300
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
over 48	8	11	17	20

4.3.2.1 Delay Element, M24, Dwg. 10519999.

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
2. Mixed ammunition	Visual
3. Flash hole missing or obstructed	Visual
4. Marking missing, incorrect or illegible	Visual
5. Delay charge not visible through flash hole	Visual
MAJOR	
101. Protective finish inadequate or defective	Visual
102. Foreign matter	Visual
MINOR	
None defined	

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.2.2 Ring, Retaining, Dwg. 8597235, covering a detail of Dwg. D10522656.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Concentricity of inside diameter with outside diameter	Gage
102. Diameter inside, max.	Gage
103. Diameter outside, min.	Gage
104. Thickness	Gage
105. Metal defective	Visual
106. Protective finish inadequate or defective	Visual
107. Burr	Visual
108. Foreign matter	Visual
MINOR	
201. Diameter, inside, min.	Gage
202. Diameter, outside, max.	Gage
203. Total length	Gage
204. Radius missing or incorrect	Visual
205. Surface finish improper	Visual

4.3.2.3 Case, Cartridge, Dwg. C10519996 covering a detail of Dwg. D10522656.

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Concentricity of head with body	Gage
102. Straightness of base	Gage
103. Diameter of head, min.	Gage
104. Thickness of wall, min.	Gage
105. Thickness of base, min.	Gage
106. Protective finish inadequate or defective	Visual
107. Burr or sharp corner	Visual
108. Foreign matter	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.2.3 (Cont'd)

	Defect	Method of Inspection
MINOR		
201.	Diameter of head, max.	Gage
202.	Diameter of body	Gage
203.	Total length	Gage
204.	Thickness of wall, max.	Gage
205.	Thickness of base, max.	Gage
206.	Location of datum diameter	Gage
207.	Radius missing or incorrect	Visual
208.	Surface finish improper	Visual

4.3.2.4 Cartridge, M174 Assembly, Dwg. D10522656.

CRITICAL		
1.	Any part missing	X-ray 1/ & Visual
2.	Improper assembly	X-ray 1/ & Visual
3.	Mixed ammunition	Visual
4.	Metal defective	Visual
5.	Marking missing, incorrect or illegible	Visual
MAJOR		
101.	Concentricity of head with body	Gage
102.	Diameter of cartridge head recess, min.	Gage
103.	Depth of cartridge head recess	Gage
104.	Diameter of body, max.	Gage
105.	Diameter of cartridge head, max.	Gage
106.	Location of datum diameter	Gage
107.	Cartridge profile failure	Gage
108.	Firing pin or retaining cup above flush with face of head	Gage
109.	Foreign matter	Visual
MINOR		
201.	Total length	Gage
202.	Fold, wrinkle, dent, buckle, bulge	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

1/ The cartridge shall be x-rayed with the long axis of the cartridge perpendicular to the rays of the x-ray machine. The x-ray negatives, properly identified, shall be forwarded to the procuring agency.

4.3.2.5 Carton, Inner, Packing, Ammunition for Cartridge Initiator, Delay, M174 Assembly, Dwg. C8864168.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual & Visual
202. Quantity of contents incorrect	Visual
203. Separator or filler missing	Visual
204. Heat sealer incomplete, loose or distorted	Visual
205. Cut or damage through impregnated box	Visual
206. Marking missing, incorrect or illegible	Visual

4.3.2.6 Box Packing for Cartridge Delay, M174, Assembly Dwg. C9214156.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual & Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of cartons incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Protective finish.- Twenty three units shall be selected for this test. Failure of any unit to comply with the requirements of 3.2 shall be cause for rejection of the lot.

4.3.3.2 Hardness.- Fifty units shall be selected for this test. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot.

4.3.3.3 Airtightness.- Each unit shall be subjected to the airtightness test. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot.

4.3.3.4 Vibration.- Twelve of the 30 assembled, x-rayed cartridges of the initial production sample, selected for ballistic tests and properly identified shall be subjected to vibration as in 3.3. After the vibration test, the 12 vibrated cartridges shall be x-rayed as directed in Note 1/ of 4.3.2.4. Evidence of any disintegration of the primer, propellant, igniter, or delay charge or derangement of any component shall be cause for rejection of the lot. When a change occurs in the lot of primers, propellant or igniters after the initiation of production, 12 assembled cartridges from the first production lot shall be tested in accordance with this paragraph.

4.3.3.5 Primer sensitivity.- A sample of 50 assembled cartridges from the initial production sample and 25 cartridges from each subsequent lot shall be tested for sensitivity. Failure of any primer to comply with the requirements of 3.4 shall be cause for rejection of the lot.

4.3.3.6 Sealant application.- Five cartridges from each production lot shall be utilized for this test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the lot.

4.3.3.7 Functional.- The following tests shall be performed by the Government:

4.3.3.7.1 Ballistics.- A sample of 30 assembled cartridges from the initial production sample (including the 12 vibrated cases) shall be ballistically tested after temperature conditioning in accordance with Table VI and 4.4.5.1. This table shall also apply if there has been any change in process or components indicated in 4.3.3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

Table VI

Temperature Degrees(o) Fahrenheit(o)	Sample
-65 to -70	10
65 to 75	10
160 to 165	10

4.3.4 Inspection equipment.- The examination and tests shall be performed using equipment listed on IEL 10522656.

4.4 Test methods and procedures.

4.4.1 Protective finish.- Test procedures for protective finish shall be in accordance with MIL-C-60539.

4.4.2 Hardness.- The hardness test shall be performed on all heat-treated components prior to the application of protective finish. A hardness testing machine shall be used to determine compliance with the drawing requirements.

4.4.3 Airtightness.- The method for determining airtightness shall be by use of the equipment listed on IEL 10522656.

4.4.4 Vibration.- The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.5 Primer sensitivity.- Remove and discard the charged delay body and all powder charges from the test cartridges by sectioning the cartridge case around its periphery 1 inch maximum above the base. Drop test 50 sectioned cartridges from the initial production sample and 25 sectioned cartridges from each production lot at 9 inches, using a steel ball 15/16 inches in diameter, weighing 1.94 ounces plus or minus 0.02 ounces in a primer sensitivity machine with a cartridge holder.

4.4.6 Sealant application.- The cartridge shall be disassembled in such a manner that determination can be made that the sealant has been applied for the full length of 360° of the joining surfaces at the junction of the cartridge head and case sidewall and that the propellant cavity is free of sealant.

4.4.7 Ballistics.- The test sample shall be rigidly supported in the fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample shall be reconditioned in accordance with 4.4.7.1.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:21Z
Check the source to verify that this is the currentversion before use.

4.4.7.1 Temperature conditioning.- Cartridges shall be loaded into acceptable M88 Initiators with end caps to cover the gas ports during the conditioning period. The temperature of the propellant within the cartridges shall be considered as the conditioning temperature and will be determined by means of a thermocoupler attached to the center of the propellant charge of a specimen cartridge assembled in a standard test device similar to the initiator undergoing tests. The conditioning time shall be 1 hour longer than the time necessary to bring the element contained in the specimen cartridge to the required temperature.

5. PREAPRATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Drawings C8864168 and C9214156 except that all markings shall reference Cartridge, Delay, M174 Assembly.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number and date of this specification.

6.2 Ignition delay is defined as the time that elapses between firing of the primer and the beginning of a continuous pressure rise on a pressure time record.

6.3 Supersession data.- This specification includes the requirements of Purchase Description FA-PD-MI-2559 dated 25 October 1962.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1377-A075

☆ U. S. GOVERNMENT PRINTING OFFICE: 1969-393-066/S-2111

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

TRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the e of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:21Z
Check the source to verify that this is the current version before use.